

Phase IV: Software Testing

Section	What to Include
Introduction	Software testing is the process of evaluating software to identify defects, bugs, or unexpected behavior before deployment. Testing helps ensure that software works correctly under different conditions and satisfies system requirements. Testing is important because it improves software reliability, correctness, security, maintainability, and overall user experience. In the Tiny Trackers daycare management system, testing ensures that parents and staff receive accurate and reliable information. The purpose of testing is to identify problems early in development and verify that all software components behave as expected under valid and invalid conditions. Testing in Tiny Trackers ensures that staff can correctly update daycare information and parents can view synchronized updates through the Parent Dashboard.
Chosen Component	The selected components for testing are Meal Management, Activity Tracking, and Sleep Tracking. These components are critical because they are used daily by daycare staff and parents. The Meal Management feature updates the daycare menu from the Staff Dashboard to both the Main Menu page and Parent Dashboard. The Activity Tracking feature records children's activities and uploads activity photos visible to parents. The Sleep Tracking feature records nap schedules and sleep duration updates

Test Cases

TC01

Valid meal update

Staff enters daily menu correctly

Meal menu is saved successfully and displayed on Main Menu and Parent Dashboard

Meal displayed correctly on both pages

Passed

TC02

Empty meal field

Staff submits empty meal form

Validation message is displayed

Validation message displayed

Passed

TC03

Meal update synchronization

Staff changes lunch menu

Updated meal appears instantly on Main Menu and Parent Dashboard

Menu updated successfully on both pages

Passed

TC04

Special characters in meal text

Staff enters symbols only (@@@@)

Error message displayed

Error message displayed

Passed

TC05

Long meal description

Staff enters very long meal text

System handles input correctly

Meal saved/displayed correctly

Passed

TC06

Extra spaces in meal input

Test Cases

TC07

Valid activity entry

Playing outside

Activity saved successfully

Activity saved successfully

Passed

TC08

Empty activity field

Empty input

Validation message displayed

Validation message displayed

Passed

TC09

Invalid activity input

@@@###

Error message displayed

Error message displayed

Passed

TC10

Long activity description

Long text over limit

System handles input correctly

Validation handled correctly

Passed

TC11

Activity photo upload

Staff uploads activity image

Photo displayed on Parent Dashboard

Photo displayed successfully

Passed

Test Cases	TC12
	Valid sleep record
	2-hour nap
	Sleep record saved successfully
	Sleep record saved successfully
	Passed
	TC13
	Empty sleep duration
	Empty field
	Validation message displayed
	Validation message displayed
	Passed
	TC14
	Negative sleep duration
	-2 hours
	Error message displayed
	Error message displayed
	Passed
	TC15
	Boundary case
	24-hour duration
	System validates correctly
	Validation handled correctly
	Passed
	TC16
	Sleep data synchronization
	Staff updates sleep schedule
	Parent Dashboard displays updated sleep information
	Sleep information updated successfully
	Passed
Testing Tools	Frontend testing the backend features, terminal testing

The Tiny Trackers tests were executed manually through the frontend interface and verified using the backend database updates and dashboard synchronization. The testing process focused on how frontend actions interacted with backend functionality.

The tests were performed by logging into the Staff Dashboard and entering data into the Meal Management, Activity Tracking, and Sleep Tracking forms. After submission, the system behavior was checked on the Main Menu page and Parent Dashboard to confirm that the backend successfully processed and synchronized the information.

Meal Management Results

The Meal Management feature was tested by entering meal information through the Staff Dashboard form.

Results:

- Valid meal updates were successfully saved in the backend database.
- Updated meals appeared correctly on:
 - Main Menu page
 - Parent dashboard menu page
- Invalid text and symbols were rejected correctly.
- Meal updates synchronized instantly between pages without requiring manual refresh.

Activity Tracking Results

The Activity Tracking feature was tested by submitting activity descriptions and uploading activity photos from the Staff Dashboard.

Results:

- Activity descriptions were saved correctly in the backend.
- Uploaded activity photos appeared successfully on the Parent Dashboard.
- Empty activity fields triggered validation errors.
- Long text and invalid inputs were handled correctly by the frontend validation system.
- Parent users could view newly uploaded activities immediately after submission.

Sleep Tracking Results

The Sleep Tracking feature was tested by entering sleep duration and nap information through the Staff Dashboard.

Results:

- Valid sleep records were saved correctly in the backend database.
- Updated sleep information appeared on the Parent Dashboard.
- Negative or empty sleep duration values triggered validation messages.
- Boundary cases were handled correctly by the system.
- Data synchronization between frontend pages worked correctly.

Execution Results

Coverage and Reflection

Test coverage is important because it ensures that the most important functionality and user interactions of the system are verified. The Tiny Trackers testing covered both frontend behavior and backend functionality.

Frontend Coverage

The frontend testing verified:

- Form input validation
- Empty field handling
- Error message display
- User interaction with forms
- Dashboard updates
- Page synchronization

Backend Coverage

The backend testing verified:

- Data storage functionality
- Database updates
- Synchronization between dashboards
- Correct retrieval of stored data
- Validation logic processing

Reflection

The testing process showed that the Tiny Trackers system functions reliably for its core features. The connection between frontend forms and backend database operations worked correctly throughout all tested scenarios.

The synchronization between:

- Staff Dashboard
- Main Menu page
- Parent Dashboard

was one of the most important parts tested because it ensures that parents always receive updated information.

Although the testing provided strong coverage for the core features, some improvements could still be made in future versions, including:

- Automated testing using Selenium or Django Test Framework
- Performance testing with many users
- Security testing for authentication and permissions
- Mobile responsiveness testing
- API and database stress testing

```
//MEALS
```

```
function saveMeals() {  
  const mealData = {  
    monday: {  
      breakfast: document.getElementById("mondayBreakfast").value,  
      lunch: document.getElementById("mondayLunch").value,  
      snack: document.getElementById("mondaySnack").value  
    },  
    tuesday: {  
      breakfast: document.getElementById("tuesdayBreakfast").value,  
      lunch: document.getElementById("tuesdayLunch").value,  
      snack: document.getElementById("tuesdaySnack").value  
    },  
    wednesday: {  
      breakfast: document.getElementById("wednesdayBreakfast").value,  
      lunch: document.getElementById("wednesdayLunch").value,  
      snack: document.getElementById("wednesdaySnack").value  
    },  
    thursday: {  
      breakfast: document.getElementById("thursdayBreakfast").value,  
      lunch: document.getElementById("thursdayLunch").value,  
      snack: document.getElementById("thursdaySnack").value  
    },  
  
    friday: {  
      breakfast: document.getElementById("fridayBreakfast").value,  
      lunch: document.getElementById("fridayLunch").value,  
      snack: document.getElementById("fridaySnack").value  
    }  
  };  
};
```



```
localStorage.setItem(
    "weeklyMeals",
    JSON.stringify(mealData)
);
document.getElementById("mealMessage").innerHTML =
    "Weekly Menu Updated Successfully!";
}
```

// ACTIVITIES

```
function saveActivity() {
    let title =
        document.getElementById("activityTitle").value;
    let description =
        document.getElementById("activityDescription").value;
    let photo =
        document.getElementById("activityPhoto").files[0];
    const reader = new FileReader();
    reader.onload = function(e) {
        let activities =
            JSON.parse(localStorage.getItem("activities")) || [];
        activities.push({
            title: title,
            description: description,
            image: e.target.result
        });
        localStorage.setItem(
            "activities",
            JSON.stringify(activities)
        );
        displayActivities();
    };
};
```

```

    if (photo) {
        reader.readAsDataURL(photo);
    }
}

function displayActivities() {
    let activities =
        JSON.parse(localStorage.getItem("activities")) || [];
    let output = "";
    activities.forEach(activity => {
        output += `
        <div class="card p-3 mb-4">
            <h4>${activity.title}</h4>
            <p>${activity.description}</p>
            
            </div>
        `;
    });
    document.getElementById("activityPreview").innerHTML =
        output;
}

displayActivities();

// SLEEP TRACKING

function saveSleepTracking() {
    let child =

```

```
    document.getElementById("sleepChild").value;

let time =

    document.getElementById("sleepTime").value;

let notes =

    document.getElementById("sleepNotes").value;

let sleepData =

    JSON.parse(localStorage.getItem("sleepTracking")) || [];

sleepData.push({

    child: child,

    time: time,

    notes: notes

});

localStorage.setItem(

    "sleepTracking",

    JSON.stringify(sleepData)

);

alert("Sleep tracking saved!");

}
```